

Question_7:

Code:

```
1  /*
2  QUESTION_7:
3  Given a linked list, delete all nodes that occur BEFORE
4  a specified node in the linked list.
5  If the given node is not present:
6  Print "Invalid Node!"
7  and display the original linked list.
8  */
9  #include <iostream>
10 using namespace std;
11
12 struct node {
13     int data;
14     node *next;
15 };
16 void create(node *&head, int x) {
17     node *p = new node{x, NULL};
18
19     if (!head)
20         head = p;
21     else {
22         node *t = head;
23         while (t->next)
24             t = t->next;
25         t->next = p;
26     }
27 }
28 int main() {
29     int n, x, p;
30     cin >> n;
31
32     node *head = NULL;
33     for (int i = 0; i < n; i++) {
34         cin >> x;
35         create(head, x);
36     }
37     cin >> p;
38     node *t = head;
```

```
39  
40     while (t && t->data != p)  
41     |     t = t->next;  
42     if (!t) {  
43     |     cout << "Invalid Node! Linked List:";  
44     |     t = head;  
45     |     while (t) {  
46     |     |     cout << "->" << t->data;  
47     |     |     t = t->next;  
48     |     }  
49     }  
50     else {  
51     |     head = t;  
52     |     cout << "Linked List:";  
53     |     while (head) {  
54     |     |     cout << "->" << head->data;  
55     |     |     head = head->next;  
56     |     }  
57     }  
58     return 0;  
59 }
```

Output:

Test Case_1:

```
4  
9 77 12 6  
12  
Linked List:->12->6  
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
4  
91 77 12 61  
2  
Invalid Node! Linked List:->91->77->12->61  
PS C:\Users\Prabin Yadav\Documents\III SEM\D
```